RAJAT BUTOLA (傅能杰)

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PROFESSIONAL SUMMARY

AI/ML Engineer with a Ph.D. in EECS specializing in building data-driven and on-premise AI systems for large-scale, time-series and high-dimensional datasets. Experienced in developing machine learning and LLM-based solutions for predictive modeling. Expertise in Python, deep learning frameworks, and production-ready AI pipelines, with a focus on scalable, interpretable, and efficient systems.

CORE COMPETENCIES & TECHNICAL SKILLS

AI / LLM / GenAI	LangChain, HuggingFace Transformers, OpenAI API, Llama, RAG Pipelines, Agentic Workflows, MCP, Prompt Engineering, Embedding models, Quantization
ML Frameworks	PyTorch, TensorFlow, Keras, Scikit-learn
ML Techniques	Supervised / Unsupervised / Semi-supervised Learning, ANN, CNN, GNN, LSTM, Transformer, Autoencoder, YOLO, VGG, ResNet, XGBoost, Random Forest, SVM
Data & Systems	Feature Engineering, Statistical modeling, Outlier Detection, Time Series Analysis, Dimensionality Reduction, FAISS, Qdrant
Languages & Tools	Python, R, MATLAB, SQL, Pandas, NumPy, FastAPI, CUDA, ReactJS, Node.js, Docker, Git, Jira
Semiconductor / EDA	Compact Modeling, Process Variation Analysis, TCAD, HSPICE, GAA/FinFET Device Simulation

PROFESSIONAL EXPERIENCE

Post-Doctoral Researcher/AI Engineer | Tzu Chi Hospital — Research Dept., Taipei, Taiwan *Dec 2025 – Present*

- Designed and implemented an AI-assisted Evidence-Based Medicine platform that automates research retrieval, screening, and structured extraction (PICO, relevance scoring) using LLM-based RAG pipelines reducing manual review time by ~60%.
- Building Transformer-based model for longitudinal EHR time-series analysis, performed feature engineering on irregularly sampled, high-dimensional clinical datasets to enable early disease-risk prediction.
- Developing an LLM-based academic progress assistant and AI coach, integrated with an institutional e-learning platform, providing students real-time, privacy-compliant visibility into course completion and graduation status.
- Optimized end-to-end data ingestion pipelines integrating SQL-backed structured storage, LLM inference, and multi-stage processing layers, resolved system bottlenecks across ingestion, database transactions, and model inference to improve pipeline robustness and throughput.

Post-Doctoral Researcher | Academia Sinica, Taipei, Taiwan *Feb 2025 – Nov 2025*

- Designed and automated end-to-end genomic data pipelines (QC → normalization → feature engineering → modeling →deployment) for cancer immunotherapy response prediction and enabling reliable ML predictions under high-dimensional, low-sample constraints.
- Trained, evaluated, and compared multiple ML classification models (XGBoost, Random Forest, Logistic Regression, SVM) to identify biomarkers for RCC and mUC cancer types from RNA-seq data, achieving AUC ~0.75 with rigorous cross-validation and feature importance analysis.
- Deployed predictive models on-premise via an interactive R Shiny application containerized with Docker, enabling reproducible, portable inference on structured biological datasets served via Shiny App Server facilitating cross-functional user access.

EDUCATION

Ph.D., Electrical Engineering & Computer Science (EECS) | National Yang-Ming Chiao Tung University, Taiwan
Sept 2018 – Dec 2023

Research: Deep Learning for Emerging GAA MOSFET Nano Device and circuit Modeling

- Analyzed large, complex nonlinear datasets from semiconductor nano-devices (GAA Nanosheet, CFET) to identify the effects caused by process variations on DC/AC/RF electrical characteristics and extracting KPI insights for design optimization and tape-out readiness.
- Developed ANN and LSTM models to detect, classify, and predict anomalous device behavior under varying process conditions (gate length, oxide thickness, work-function fluctuation), reducing TCAD simulation time by >40% while maintaining >95% predictive accuracy.
- Applied CNN and dimensionality reduction (DR) to identify problematic OPC mask patterns, accelerating lithography correction workflows while maintaining prediction accuracy.

- Developed compact neural network models as drop-in replacements for TCAD/HSPICE simulators using hybrid TCAD+SPICE datasets and semi-supervised training techniques to minimize labeled data requirements and improve scalability.
- Collaborated with cross-functional teams (device, circuit, and fabrication) and published findings in top-tier IEEE journals and international conferences.

Master, Electronics & Communication Engineering | Jaypee Institute of Information Technology University, India

Aug 2010 – May 2012

- Dissertation: Image Enhancement using Thresholding & Histogram Equalization (HE), built full evaluation pipeline for brightness-preserving HE frameworks using MATLAB and Python.

KEY PROJECTS

On-Premise AI-Assisted Systematic Review Platform

- Built a fully on-premise LLM-powered platform for evidence-based medicine using open-source models, RAG pipelines, and FAISS/Qdrant vector search (no cloud API dependency). Automated multi-stage data ingestion, relevance classification, and structured extraction with SQL-backed storage and a Streamlit web interface supporting multi-reviewer blinded screening, PRISMA diagram generation, and parallel full-text retrieval across different databases.

ImmunoResponse Predictor

- Developed ML pipeline for multi-cancer biomarker identification and treatment response prediction, deployed via Docker-containerized R Shiny App Server enabling reproducible, team-accessible inference without cloud infrastructure.

DW-ANN Compact Model for Circuit Simulation

- Designed a novel Dynamic Weighting ANN (DW-ANN) trained on hybrid TCAD/SPICE datasets to predict device performance anomalies across process corners, achieved <2% error on inverter, SRAM, and ring oscillator circuits, reducing training data generation cost by >50%.

PUBLICATIONS

Journals:

1. **R. Butola**, Y. Li, and S. R. Kola "Compact Model Build upon Dynamic Weighting Artificial Neural Network Approach for Complementary Field Effect Transistors," IEEE Transactions on Electronic Devices (IEEE-TED), vol. 71, Jan. 2024.
2. S. R. Kola, Y. Li, and **R. Butola**, "Statistical Device Simulation and Machine Learning of Process Variation Effects of Vertically Stacked Gate-All-Around Si Nanosheet CFETs," in IEEE Transactions on Nanotechnology, vol. 23, April, 2024.
3. **R R. Butola**, Y. Li, and S. R. Kola "Application of Long Short-Term Memory Modeling Technique to Predict Process Variation Effects of Stacked Gate-All-Around Si Nanosheet Complementary-Field Effect Transistors," Computers & Electrical Engineering, Elsevier, vol. 105, Jan. 2023.
4. **R. Butola**, Y. Li, and S. R. Kola "A Comprehensive Technique based on Machine Learning for Device and Circuit Modeling of Gate-All-Around Nanosheet Transistors," IEEE OJ-Nano, vol. 4, Oct. 2023.
5. **R. Butola**, Y. Li, and S. R. Kola, C. Y. Chen, and M-H. Chuang "Artificial Neural Network-Based Modeling for Estimating the Effects of Various Random Fluctuations on DC/Analog/RF Characteristics of GAA Si Nanosheet FETs," IEEE Trans. Microwave Theory & Techniques, vol. 70, no. 11, Nov. 2022.
6. **R. Butola**, Y. Li, and S. R. Kola "A Machine Learning Approach to Modeling Intrinsic Parameter Fluctuation of GAA Si Nanosheet MOSFETs," IEEE Access, vol. 10, July 2022.
7. P. Parashar, C. Akbar, T. S. Rawat, S. Pratik, **R. Butola**, S. H. Chen, Y-S Chang, S. Nuannimnoi, and A. S. Lin "Intelligent Photolithography Corrections Using Dimensionality Reductions," IEEE Photonics Journal, vol. 11, no. 5, Oct. 2019.
8. P. Parashar, C. H. Chen, C. Akbar, S. M. Fu, T. S. Rawat, S. Pratik, **R. Butola**, S. H. Chen, A. S. Lin. "Analytics-statistics mixed training and its fitness to semi-supervised manufacturing" PLoS ONE Journal, 2019.

Conferences:

1. **R. Butola**, Y. Li, and S. R. Kola "A DNN-Based Compact Modeling Technique for GAA Si NS FETs and Its Application in CMOS Circuit Simulation," International conferences on the science and technology of devices and materials (SSDM), Japan, 2023.
2. **R. Butola**, Y. Li, and S. R. Kola, M-H. Chuang, and C. Akbar "Estimating the Process Variation Effects of Stacked Gate All Around Si Nanosheet CFETs Using Artificial Neural Network Modeling Framework," IEEE International Conference on Nanotechnology (IEEE-NANO), Spain, 2022.
3. **R. Butola**, Y. Li, and S. R. Kola "Deep Learning Approach to Estimating Work Function Fluctuation of Gate-All-Around Silicon Nanosheet MOSFETs with A Ferroelectric HZO Layer," IEEE-EDTM, Japan, 2022.
4. **R. Butola**, Y. Li, and S. R. Kola "Machine Learning Approach to Characteristic Fluctuation of Bulk FinFETs Induced by Random Interface Traps," IEEE-ISQED, USA, 2022.
5. **R. Butola**, Y. Li, and S. R. Kola "Deep Learning Approach to Modeling and Exploring Random Sources of Gate-All-Around Silicon Nanosheet MOSFETs," International Symposium on VLSI Technology, Systems and Applications (VLSI-TSA), Hsinchu, Taiwan, 2022.